Generative AI Internship

Task 03: Text Generation with Markov Chain

Submitted By:

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Internship:

Generative AI Virtual Internship

Organization:

PRODIGY InfoTech

1. Introduction:

Text Generation is an important application of Natural Language Processing (NLP) and Generative Artificial Intelligence. In this project, a Markov Chain model is used to generate meaningful text by learning word relationships from a given dataset. The model predicts the next word based on probability and produces new sentences that resemble the original text.

2. Objective:

The objective of this project is to implement text generation using the Markov Chain algorithm and understand how probabilistic language models generate meaningful text from a given dataset.

3. Tools and Technologies Used:

* Python

* Google Colab

* Markovify

* Jupyter Notebook

* GitHub

4. Project Workflow:

Step 1

Open Google Colab and install the Markovify library.

Step 2

Import the required Python modules.

Step 3

Create a sample dataset containing AI-related text.

Step 4

Train the Markov Chain model using the dataset.

Step 5

Generate multiple sentences using the trained model.

Step 6

Display and verify the generated output.

5. Implementation:

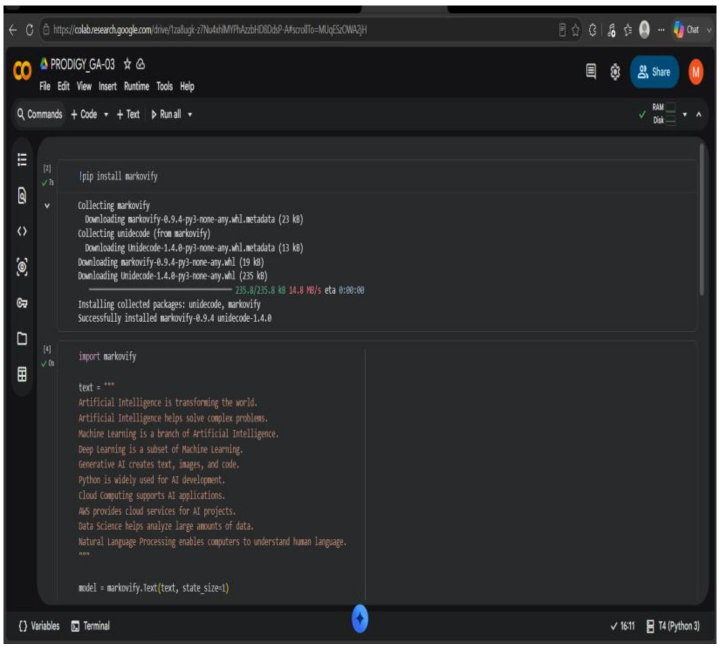
The Markovify library was used to build a Markov Chain model from a sample text dataset. The model analyzed word transitions and generated new sentences by predicting the most probable next word. Multiple sentences were generated to evaluate the quality of the output.

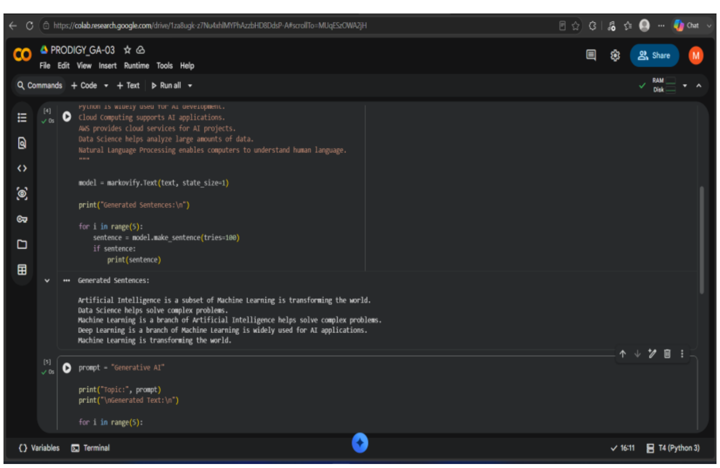
6. Results:

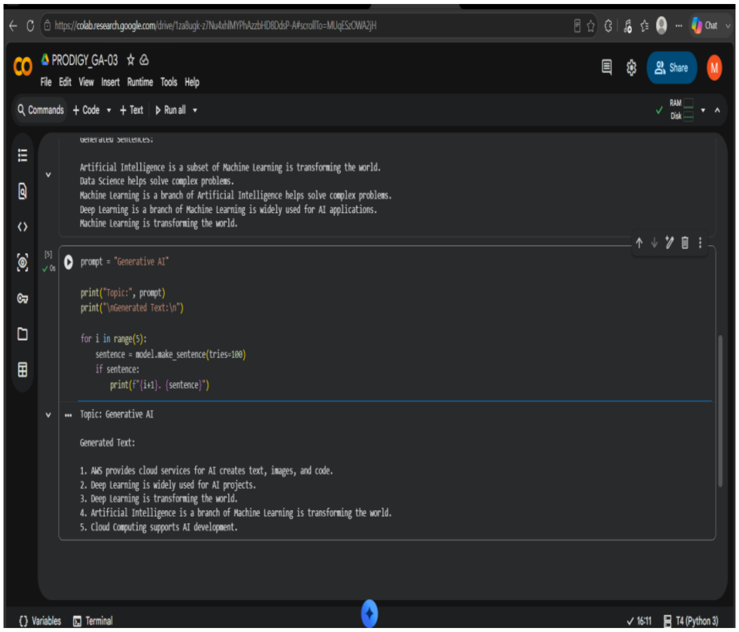
The Markov Chain model successfully generated readable and meaningful text based on the provided dataset. The generated sentences demonstrated the working of probabilistic text generation and basic Natural Language Processing concepts.

7. Output:

The notebook successfully generated multiple text sentences using the Markov Chain algorithm. Screenshots of the code execution and generated outputs are included in the GitHub repository.







8. Learning Outcomes:

* Learned the fundamentals of Markov Chains.

* Understood probabilistic text generation.

* Implemented a simple NLP project using Python.

* Gained hands-on experience with the Markovify library.

* Improved practical knowledge of Generative AI concepts.

9. Conclusion:

This project successfully demonstrated text generation using the Markov Chain algorithm. It provided practical experience with probabilistic language models and strengthened the understanding of text generation techniques used in Artificial Intelligence.

GitHub Repository Contents:

* PRODIGY_GA_03.ipynb – Google Colab Notebook

* README.md – Project Description

* Outputs – Execution Screenshots

* Final_Documentation.pdf – Project Report

Author:

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